

MODERNA

Case Study

Biotechnology Innovation: mRNA
Technology

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About Moderna

- Founded in 2010
- Headquarters: Cambridge, Massachusetts, USA
- Develops mRNA medicines and vaccines

History

- Started as an mRNA research company
- Global breakthrough with COVID-19 vaccine

mRNA Technology

- Uses messenger RNA to instruct cells to produce proteins
- Triggers immune response
- Faster than conventional vaccine development

Products

- Spikevax (COVID-19)
- RSV vaccine
- Personalized cancer vaccines under development

Applications

- Infectious diseases
- Cancer immunotherapy
- Rare diseases

Advantages

- Rapid development
- High efficacy
- Flexible platform

Challenges

- Cold storage
- High development costs
- Regulatory approval

Market Impact

- Billions of vaccine doses supplied
- Expanded mRNA research worldwide

Future Scope

- Personalized medicine
- Combination vaccines
- New mRNA therapies

Conclusion

Moderna has revolutionized biotechnology using mRNA technology and continues to drive innovation.

References

- Moderna
- WHO
- NIH
- Nature Reviews Drug Discovery